CALVIN KRANIG

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EDUCATION

BS Iowa State University, Computer Science December 2019
Graduated Summa Cum Laude with Honors

MS University of Wisconsin Madison, Computer Science May 2024

PHD University of Wisconsin Madison, Computer Science Projected May 2027

HONORS AND AWARDS

Suma Cum Laude, Iowa State University 2019
Graduated with a 4.0 and was in the top 2% of all graduates

Distinguished Military Graduate, U.S. Army 2019
Awarded to top 20% of ROTC Cadets on the National Order of Merit List

Arthur A. Collins Scholarship, Iowa State University 2018
Awarded to students with 3.0+ GPA who have taken leadership positions on campus

Joseph Cribbins Scholarship, Civilian Sponsored Scholarship Program 2018
Awarded to top 2 ROTC Cadets in the nation with STEM majors

MS2 Cadet of the Year, Iowa State University 2017
Awarded to the top 2nd year Cadet in the Iowa State ROTC program

RESEARCH EXPERIENCE

University of Wisconsin Madison August 2022 to May 2023
Research Assistant

- Conducted software vulnerability assessment of 3rd party IDM/Authentication software
- Developed software to actively query DNS for typo squatting study
- Created custom dataset, and trained reward model used in RL for soccer simulation
- Lead on study investigating DNS measurement mechanisms and datasets
- Lead on study stress testing DNS software, hardware, and networking systems

TEACHING EXPERIENCE

Iowa State University August 2019 to December 2019
Teaching Assistant, Computer Science Orientation

- Gave large lectures as needed, and led weekly small group breakouts
- 200+ undergraduate students per lecture and 30 students per small group

Iowa State University August 2016 to May 2018
President, IgniteCS

- Developed project based coursework for teaching high school students Python
- Led weekly classes containing 5-10 students

Minnesota STEM Partnership, November 2020 – February 2021
Team Coach, First Lego League,

- Developed project based coursework for teaching high school students Python
- Led weekly classes containing 5-10 students

PUBLISHED WORK

Kranig, Calvin, Eric Pauley, Wei-Shiang Wung, Paul Barford, Mark Crovella, and Joel Sommers. "Toward a Representative DNS Data Corpus: A Longitudinal Comparison of Collection Methods." In *2025 9th Network Traffic Measurement and Analysis Conference (TMA)*, pp. 1-11. IEEE, 2025.

Wung, Wei-Shiang, Calvin Kranig, Eric Pauley, Paul Barford, Mark Crovella, and Joel Sommers. "Squatspotting: Towards the Systematic Measurement of Typosquatting Techniques." IFIP Networking, 2025.

Nibbelink, Ben, Calvin Kranig, Kay Avila, Elisa Heymann, and Barton P. Miller. "Trusted CI: The NSF Cybersecurity Center of Excellence Custos First Principles Vulnerability Assessment." (2022).

PRESENTATIONS

Kranig, C. (2019). Python for High Schoolers. Project Presentation given at the bi-annual Iowa State Honors Project Presentation Ceremony

PROFESSIONAL EXPERIENCE

General Dynamics Mission Systems, Bloomington, MN
Software Engineer, January 2020 – Present

- Developed a mission critical C++ module for running inference on TensorRT optimized DNN models
- Enabled the training of a specialized contract specified object detection DNN by developing image generation scripts utilizing Blender

- Tested end product for optimal power usage and object detection performance
- Designed, developed, and programmed a virtual environment for testing drone swarm behaviors
- Developed and tested a Deep Q Learning model for handling drone swarms in a combat environment
- Created back-end interface for utilizing proprietary Air Force software for modeling and simulation
- Designed process for logging and consolidating Windows events for government level PAAS
- Provided cyber security testing and analysis for PaaS utilized by multiple government customers

Minnesota Army National Guard, Rosemount, MN
Cyber Operations Officer, June 2015 – Present

- Led and organized multiple fundraising events consisting of 100+ cadets
- Lead cyber protection team mission element consisting of 6-10 analysts
- Trained and conducted defensive cyber operations including network monitoring and malware discovery
- Trained and performed offensive cyber operations including initial exploit and privilege escalation
  Lead mission element through quarterly cyber range exercises conducting incident response

Iowa State University, Ames, IA
IT Specialist, November 2016 – December 2019

- Updated and maintained websites for the Political Science and History departments
- Maintained and upgraded local hardware and network infrastructure

National Reconnaissance Office, Chantilly, VA
Software Development Intern, Summer 2018, 2019

- Designed software architecture and logical data model for a new Mission Assurance software tool
- Applied knowledge of model based systems engineering to create a workflow model and SLA
- Advised the head of the modeling department on model development practices and procedures

LEADERSHIP AND VOLUNTEER WORK

RoboCup, University of Wisconsin Madison
Team Member, January 2023 – Present

First Lego League, Minnesota STEM Partnership
Team Coach, November 2020 – February 2021

Iowa State Honors Student Board, Iowa State University

Web Master, August 2018 – May 2019
Upper Honors Activity Planning Board Member, January 2017 - May 2018

COMPUTER SKILLS AND CERTIFICATIONS

Programming Languages: Java, Python, C, C++, MATLAB, Go

Libraries & Applications: Security Onion, Pytorch, TensorFlow, TensorRT, GRR, Wireshark, Blender, ELK Stack

Certifications: CISSP, GCLD

Clearance: TS/SCI